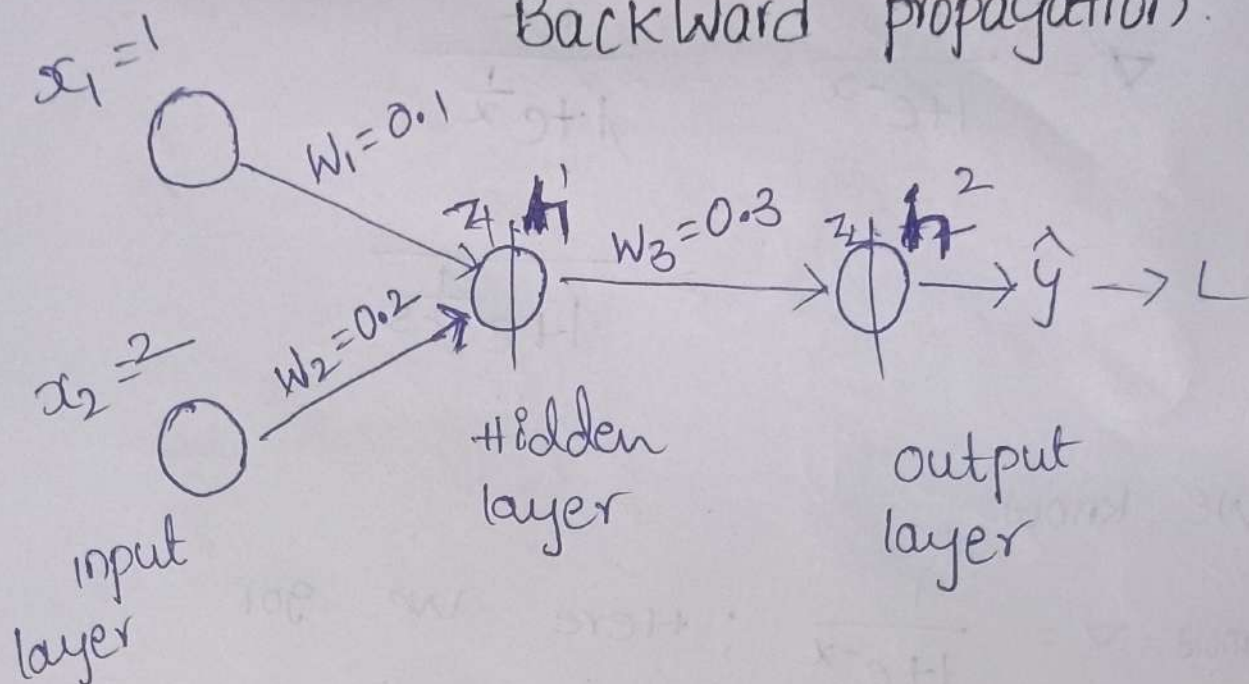


Forward propagation & Backward propagation.



Given

$$x_1 = 1$$

$$x_2 = 2$$

$$y_{act} = 1$$

$$w_1 = 0.1$$

$$w_2 = 0.2$$

$$w_3 = 0.3$$

$$\eta =$$

learning

$$rate = 0.1$$

Forward propagation

① compute forward pass & predict output?

Given

→ Activation $\Rightarrow$ Sigmoid.

→ No bias.

$$z_1' = w_1 x_1 + w_2 x_2$$

$$\therefore \phi_1 = \text{Act}(w_1 x_1 + w_2 x_2)$$

$$= \text{Act}(0.1 \times 1 + 0.2 \times 2)$$

$$= \text{Act}(0.1 + 0.4)$$

$$= \text{Act}(0.5)$$

$$; z_1 = 0.5$$

We know

$$\text{Sigmoid} = \sigma = \frac{1}{1+e^{-x}} ; \text{ Here we got}$$

$$x = 0.5$$

$$= \frac{1}{1+e^{-0.5}} = \frac{1}{1+0.606}$$

$$= \frac{1}{1.606} = 0.622$$

$$\cancel{h_1} = \cancel{0.622} \quad h' = 0.622$$

For final layer: $z^2 = (h' \times w_3)$

$$h^2 = \text{Act}(h' \times w_3)$$

$$= \text{Act}(0.622 \times 0.3)$$

$$= \text{Act}(0.1866)$$

$$= \sigma(0.1866)$$

$$z^2 = 0.1866$$

$$\Delta = \frac{1}{1+e^{-x}}$$

here we got

$$x = 0.1866$$

$$= \frac{1}{1+e^{-0.1866}}$$

$$= \frac{1}{1+0.8297}$$

$$= \frac{1}{1.8297}$$

$$h^2 = 0.5465$$

output $\Rightarrow \hat{y} = 0.5465$

②

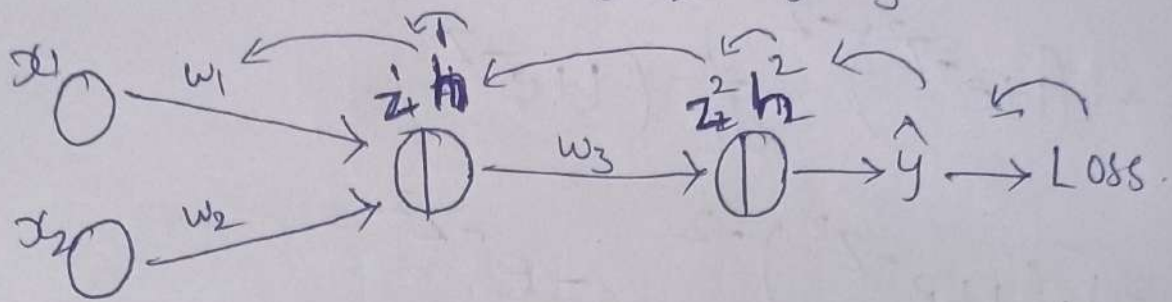
$$\text{loss} = (y - \hat{y})^2$$

$$= (1 - 0.5465)^2$$

$$= (0.4535)^2 = 0.2056$$

Backward Propagation

③ Compute $\frac{\partial L}{\partial w_1}$, $\frac{\partial L}{\partial w_2}$, $\frac{\partial L}{\partial w_3}$



$\frac{\partial L}{\partial w_1}$ → we use chain rule.

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial h^1} \cdot \frac{\partial h^1}{\partial z^1} \cdot \frac{\partial z^1}{\partial w_1}$$

$$= 2 \cdot (y - \hat{y}) \cdot (-1) \cdot 1 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2))$$

$$w_3 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2)) \cdot x_1$$

here how

$$\frac{\partial h^2}{\partial z^2} = \sigma(z^2) \cdot (1 - \sigma(z^2))$$

let's derive.

we know

$$h^2 = \sigma(z^2)$$

$$= \frac{1}{1 + e^{-z^2}}$$

$$\begin{aligned}
 \frac{dh^2}{dz^2} \left(\frac{1}{1+e^{-z^2}} \right) &= (1+e^{-z^2})^{-1} \\
 &= -1 (1+e^{-z^2})^{-2} \cdot \frac{d}{dz^2} (1+e^{-z^2}) \\
 &= -1 (1+e^{-z^2})^{-2} \cdot e^{-z^2} \\
 &= (1+e^{-z^2})^{-2} \cdot e^{-z^2} \\
 &= \frac{e^{-z^2}}{1+e^{-z^2}}
 \end{aligned}$$

This can be written as.

We know

$$\sigma(z^2) = \frac{1}{1+e^{-z^2}}$$

So

$$1 - \sigma(z^2) = \frac{e^{-z^2}}{1+e^{-z^2}}$$

Then

$$\sigma(z^2) \cdot (1 - \sigma(z^2)) = \frac{e^{-z^2}}{1+e^{-z^2}}$$

[* Same for $d\hbar'/dz'$]

$$\frac{\partial L}{\partial w_1} = 2(y - \hat{y})(-1) \cdot 1 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2))$$

$$w_3 \cdot \sigma(z') (1 - \sigma(z')) \cdot x_1$$

$$= 2(1 - 0.5465)(-1) \cdot 1 \cdot 0.5465$$

$$(1 - 0.5465) \cdot (0.3) \cdot 0.622$$

$$(1 - 0.622) \cdot 1$$

$$= 2 \cdot 0.4535 \cdot -1 \cdot 1 \cdot 0.5465 \cdot 0.4535 \cdot (0.3)$$

$$0.622 \cdot 0.378$$

$$= -0.015$$

$$\frac{\partial L}{\partial w_2} = 2(y - \hat{y})(-1) \cdot 1 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2)) \cdot$$

$$w_3 \cdot \sigma(z') (1 - \sigma(z')) \cdot x_2$$

$$= 2 \cdot 0.4535 \cdot -1 \cdot 1 \cdot 0.54$$

$$\frac{\partial L}{\partial w_1} = 2(y - \hat{y})(-1) \cdot 1 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2))$$

$$w_3 \cdot \sigma(z') \cdot (1 - \sigma(z')) \cdot x_1$$

$$= 2(1 - 0.5465)(-1) \cdot 1 \cdot 0.5465$$

$$(1 - 0.5465)(0.3) \cdot 0.622 \cdot (1 - 0.622) \cdot 1$$

$$= 2 \cdot 0.4535 \cdot -1 \cdot 1 \cdot 0.5465 \cdot 0.4535(0.3)$$

$$0.622 \cdot 0.378$$

$$= -0.015$$

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial h'} \cdot \frac{\partial h'}{\partial z'} \cdot \frac{\partial z'}{\partial w_2}$$

$$= 2(y - \hat{y})(-1) \cdot 1 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2))$$

$$w_3 \cdot \sigma(z') \cdot (1 - \sigma(z')) \cdot x_2$$

$$= 2(1 - 0.5456)(-1) \cdot 1 \cdot 0.5465 \cdot (1 - 0.5465)$$

$$(0.3) \cdot 0.622 \cdot (1 - 0.622) \cdot 0.2$$

$$= -0.003$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h^2} \cdot \frac{\partial h^2}{\partial z^2} \cdot \frac{\partial z^2}{\partial w_3}$$

$$= 2(y - \hat{y})(-1) \cdot 1 \cdot \sigma(z^2) \cdot (1 - \sigma(z^2)) \cdot h^1$$

$$= 2(1 - 0.5456) \cdot (-1) \cdot 1 \cdot 0.5465 \cdot (1 - 0.5465)$$

$$\therefore -0.622$$

$$= 2 \cdot 0.4535 \cdot (-1) \cdot 1 \cdot 0.5465 \cdot 0.4535 \cdot 0.622$$

$$\approx -0.1398$$

Now Let's compute new weights $\rightarrow$ update gradient descent rule

$$w_{\text{new}} = w_{\text{old}} - \eta \cdot \frac{\partial L}{\partial w_1}$$

$$= 0.1 - 0.1 \times -0.15$$

$$= 0.1 + 0.015$$

$$= 0.115$$

$$w_{new}^2 = w_{old}^2 - \eta \cdot \frac{\partial L}{\partial w_2}$$

$$= 0.2 - 0.1 \times -0.003$$

$$= 0.2 + 0.0003$$

$$= 0.2003$$

$$w_{new}^3 = w_{old}^3 - \eta \cdot \frac{\partial L}{\partial w_3}$$

$$= 0.3 - 0.1 \times -0.1396$$

$$= 0.3 + 0.0139$$

$$= 0.3139$$